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Personal Summary

MSc Data Science graduate with research experience spanning machine learning, climate modelling and scientific computing. Skilled in Python, PyTorch, SQL, statistical analysis and time-series modelling, with hands-on experience building reproducible research pipelines and AWS cloud systems. Combines rigorous experimental design, uncertainty quantification and clear communication to investigate complex, data-driven problems.

Education

University of Bristol

Sep 2025 – Expected Sep 2026

MSc Data Science

Bristol, United Kingdom

- Relevant study: Statistical Computing and Empirical Methods; Programming and Algorithms; AI and Text Analytics; Large-Scale Data Engineering; Data Science Methods and Practice; Visual Analytics.

Veermata Jijabai Technological Institute (VJTI)

Dec 2021 – Jun 2025

B.Tech. Production Engineering, First Class, CGPA: 7.33/10

Mumbai, India

- Relevant study: Probability and Statistics; Operations Research; Business Analytics; Programming and Problem Solving in C++; Supply Chain Management.

Publication

Bhongade, A. (2025). *The Unequal Impact of Lethal Heatwaves in India for Males and Females: Insights from Climate Projection Data*. *Cambridge Journal of Climate Research*, 2(1), 182–194. DOI: [10.60866/CAM.241](https://doi.org/10.60866/CAM.241)

- Sole-authored a peer-reviewed study integrating CMIP6 climate and demographic projections across five SSP scenarios to analyse heat exposure, cooling-energy demand, uncertainty, and age- and gender-based disparities in India through 2100.

Research Experience

University of Bristol

June 2026 – August 2026

MSc Research Dissertation – Length Generalisation in Transformer Sequence Models

Bristol, United Kingdom

- Developed an evidence-based failure taxonomy distinguishing training underfitting, seed instability, immediate extrapolation collapse and position-dependent error patterns, supported by diagnostic analyses and mitigation outcomes.
- Designed and executed 45 Transformer baseline experiments across 3 algorithmic tasks, 5 positional encodings and 3 random seeds using PyTorch and AWS GPU infrastructure, evaluating task lengths up to 1,024.
- Established a multi-seed baseline-eligibility framework, identifying 5 of 15 configurations with 92–100% mean training-length exact-match accuracy; all five reached 0% at the first tested unseen length.
- Characterised task-specific failure patterns using position-wise error profiles, attention statistics, carry-condition analysis and dependency-distance diagnostics, moving beyond aggregate accuracy.
- Evaluated mixed-length training, curriculum learning and randomised padding, separating training underfitting from interventions that preserved training performance but produced no observed improvement in unseen-length exact match.

University of Cambridge

Sep 2024 – Mar 2025

Visiting Researcher – Cambridge Collective Intelligence and Design Group

Cambridge, United Kingdom

- Built a Python pipeline integrating World Bank CMIP6 climate and demographic projections across five SSP scenarios with appliance-sales data to analyse heat exposure and cooling-energy demand in India through 2100.
- Evaluated two genders and 20 five-year age groups across four future periods, generating 40 scenario-specific contour plots alongside heat-index trends, percentile bounds, and Cooling Degree Day correlation analyses.
- Identified a rise in heat index to approximately 89°F by 2099 under SSP5-8.5, with higher cooling demand among older age groups and substantially wider uncertainty under high-emission scenarios; translated the work into a single-author peer-reviewed publication.

- Processed FITS/HEALPix weak-lensing convergence maps on the UZH Science Cluster across six simulations, five redshifts ($z = 0.6\text{--}2.0$), and four Newtonian/relativistic configurations, covering 120 configuration-redshift combinations.
- Developed Python workflows using Astropy, Healpy/Anafast, NumPy, and Matplotlib to compute angular power spectra and produce analysis-ready map statistics and visualisations for downstream research.
- Compared Newtonian and relativistic outputs using mean and relative-difference calculations, 20-bin aggregation, standard deviations, and error bars across simulation runs.

- Collaborated to identify the optimal EV charger using Multiple Criteria Decision Making (MCDM), applying MCDM frame-works and Super Decision Software to structure complex multi-variable evaluation, involving industry expert consultation, manufacturing plant visits, and extensive literature review.

Technical Projects

Elastic Event-Driven Data Pipeline on AWS

EC2, S3, SQS, DynamoDB, IAM, CloudWatch, Auto Scaling

- Deployed an event-driven processing pipeline using S3, SQS, Ubuntu EC2 workers, and DynamoDB, with least-privilege IAM roles and queue-depth scaling through CloudWatch alarms.
- Load-tested the architecture with a 120-file burst; it scaled from one to two workers, processed the complete workload without data loss, and returned to minimum capacity after the queue emptied.

Robust Prediction Intervals Under Distribution Shift

Research Proposal

- Designed a controlled evaluation of four probabilistic forecasters with native and conformal calibration under four temporal shift types and three severity levels, using coverage, interval width, CRPS, and Winkler score.

Robust Time-Series Anomaly Detection Under Instrument Shift

Python, Lightkurve, Astropy, scikit-learn

(Ongoing)

- Developing a reproducible pipeline for Kepler and TESS light curves using quality-flag filtering, missing-data handling, detrending, normalisation, and rule-based signal screening.
- Building Isolation Forest anomaly scoring for noisy astronomical observations and designing controlled comparisons against robust statistical baselines.
- Designing cross-quarter and cross-instrument evaluation to distinguish rare temporal behaviour from detector-specific distribution shift using precision-recall and false-positive analysis.

Technical Skills

Programming: Python, C++, R, SQL, Git, Linux, Jupyter, LaTeX

Machine Learning: PyTorch, TensorFlow, scikit-learn, Transformers, neural networks, regression, clustering, anomaly detection, time-series analysis, model evaluation, distribution shift, uncertainty quantification

Scientific Computing: NumPy, SciPy, pandas, Matplotlib, xarray, netCDF4, Astropy, Lightkurve, Healpy, Anafast, FITS, HEALPix

Cloud and Infrastructure: AWS EC2, S3, SQS, DynamoDB, IAM, CloudWatch, Auto Scaling, AWS CLI; SLURM and HPC workflows

Research Methods: Experimental design, statistical analysis, reproducible pipelines, ablation studies, error analysis, scientific visualisation

Leadership, Awards and Training

- AWS Academy Cloud Foundations; HarvardX R Basics; CBSE Certificate of Merit for ranking in the top 0.1% nationally in Social Science.
- **Department Manager, Enthusia VJTI:** led volunteer logistics for 500+ participants.
- **Public Relations Executive, Technovanza VJTI:** supported engagement for a national technical festival with 60,000+ attendees.
- **University Basketball Team Captain:** captained the team across seven matches, securing three inter-college wins.